

Cell Structure, Cell Types and Levels of Organisation

The parts of a cell, how plant and animal cells differ, and how cells build up into a whole organism.

Inside a cell

All living beings are made of **cells**. A typical cell has three basic parts:

Part	Function
Cell membrane	Outer cover; separates one cell from another; porous — lets useful materials in and waste out.
Cytoplasm	Jelly between membrane and nucleus; holds proteins, fats, salts; most life processes happen here.
Nucleus	Round structure that controls all activities of the cell and regulates growth.

Plant cell vs animal cell

Feature	Plant cell	Animal cell
Cell wall	Present (gives rigidity & strength)	Absent
Plastids (e.g. chloroplast)	Present (green ones do photosynthesis)	Absent
Vacuole	One large vacuole	Small or absent

Onion vs cheek: onion-peel cells (Activity 2.2) are rectangular with a cell wall; human cheek cells (Activity 2.3) are polygon-shaped with no cell wall. Stains used: **safranin** (red) for onion, **methylene blue** for cheek.

Shape follows function

- **Nerve cell** — long and branched, to carry messages quickly to distant parts.
- **Muscle cell** — spindle-shaped and flexible, to contract and relax (e.g. push food down the food pipe).

Unicellular vs multicellular

Unicellular = one cell does everything (bacteria, Amoeba). **Multicellular** = many cells with special jobs (plants, animals). A bacterium has no proper nucleus — only a **nucleoid**.

Levels of organisation

Cell → Tissue → Organ → Organ system → Organism

Similar cells form a **tissue**; tissues form an **organ**; organs working together form an **organ system**; all systems together make the **organism**.

Where students lose marks

Trap 1 — Cell wall vs cell membrane. All cells have a membrane. Only plant, fungal and bacterial cells have an extra **cell wall**. Animal cells have no cell wall.

Trap 2 — Bacteria have no true nucleus. They have a **nucleoid** (no nuclear membrane), unlike plant, animal, fungal and protozoan cells.

Trap 3 — Stains. Onion peel → safranin (red); cheek cells → methylene blue. Don't swap them. Glycerin is added to stop drying.

Order slip: levels go cell → tissue → organ → organ system → organism. Don't reverse them.

Model answers — write them like this

Example A. Give two differences between a plant cell and an animal cell.

Step 1: Cell wall: present in plant, absent in animal.

Step 2: Chloroplast & large vacuole: present in plant, absent/small in animal.

∴ Plant cells have a cell wall and chloroplasts; animal cells do not.

Example B. How does the shape of a nerve cell suit its function?

Step 1: Its job is to carry messages around the body.

Step 2: Being long and branched lets it reach distant parts quickly.

∴ The long, branched shape helps it pass messages fast.

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